

**BANARAS HINDU UNIVERSITY**

B.Voc. Semester III Examination, 2022-23

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**Paper: BVCA-302 — Robotics with Arduino**

Time: Three Hours

Maximum Marks: 70

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*Instructions: Answer all Sections. Candidates are required to write their answers in their own words as far as practicable.*

**SECTION — A: Long Answer Questions**

[15 × 2 = 30]

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*Instructions: Long answer type questions to be answered in about 750 words each. Attempt any two.*

**Question 1.** Discuss the different types of Robots and elaborate on their functionality in detail.

**OR**

Discuss the different types of Sensors along with their applications and functionality.

**Question 2.** Draw the block diagram of a microcontroller and explain each of its internal components.

**OR**

What are the major steps required to install Proteus simulation software? Discuss the various features available in it.

**SECTION — B: Short Answer Questions**

[5 × 6 = 30]

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*Instructions: Short answer type questions to be answered in about 450-550 words each. Attempt any six.*

**Question 3.** Discuss in detail the functions and the need for Industrial Robots in manufacturing.

**Question 4.** What are the major physical and structural components of a Robot? Explain.

**Question 5.** List some of the important reasons for using robots instead of humans to perform a task.

**Question 6.** What are the advantages of integrating Artificial Intelligence (AI) into modern Robotics?

**Question 7.** Discuss different types of Robot Locomotion. Explain their working principles in detail.

**Question 8.** What are the differences between voice recognition and speech recognition?

**Question 9.** What is a pressure sensor? List any two advantages of pressure sensors in robotics.

**Question 10.** What is a servo motor? List any two applications of servo motors.

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**SECTION — C: Multiple Choice Questions**

[10 × 1 = 10]

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*Instructions: Select the correct option for each question.*

**Question 1.** The main objective(s) of an Industrial robot is/are to:

- (a) Minimize the labor requirement
- (b) Increase productivity
- (c) Enhance the life of production machines
- (d) All of the above

**Question 2.** Which of the following branches processes sensor feedback in robotics?

- (a) Computer Engineering
- (b) Mechanical Engineering
- (c) Electrical Engineering
- (d) Electronic Engineering

**Question 3.** Which of the following sensors works based on radio detecting and ranging?

- (a) Sonar
- (b) Radar
- (c) Inertial
- (d) Biosensor

**Question 4.** Why do robots need sensors?

- (a) To collect information from the environment
- (b) To map environment attributes to a quantitative measurement
- (c) Only option (a) is true
- (d) Both options (a) and (b) are true

**Question 5.** Which sensor can be used in robots for measuring distance?

- (a) MPU6050
- (b) LDR module
- (c) Ultrasonic Sensor
- (d) None of the above

**Question 6.** A Robot is:

- (a) Programmable

- (b) A multifunctional manipulator
- (c) Both (a) and (b)
- (d) None of the above

**Question 7.** Drives are also known as:

- (a) Actuators
- (b) Controllers
- (c) Sensors
- (d) Manipulators

**Question 8.** If a robot has  $k$  legs, then the number of possible legged events is represented by:

- (a)  $N = 2k - 2$
- (b)  $N = (2k - 2)!$
- (c)  $N = (2k - 1)!$
- (d)  $N = (2k)!$

**Question 9.** Which of the following is not an essential component for the construction of robots?

- (a) Power Supply
- (b) Actuators
- (c) Sensors
- (d) Energy

**Question 10.** Which of the following is not a type of Robot Locomotion?

- (a) Legged
- (b) Wheeled
- (c) Tracked slip
- (d) Tracked skid